

Why Rotating Fluids Make Polygons

Paper III: Gravitational Theory

Gordon Speagle

Abstract

Building on the vortex–gravity isomorphism established in Paper II, we develop the gravitational theory of polygon stability in 2+1 and higher dimensions. The orbifold CFT at central charge $c = 12b(N) \approx N^2$ identifies the Havelock eigenvalue with the twisted-sector ground-state energy (the orbifold–Havelock correspondence). The polygon–BTZ phase transition has critical exponent $\nu = 1/2$, with entropy decomposing into discrete (wall-crossing, $\log 2$ per threshold) and continuous (one-loop) parts. The Wheeler–DeWitt equation on the palindromic minisuperspace connects the stability threshold to the Hamiltonian constraint. The main result: in 2+1D, the Havelock eigenvalue structure—applied via the equivalence principle (test polygons at every point of a spatial slice)—*derives* the vacuum Einstein equations $R_{\mu\nu} = \Lambda g_{\mu\nu}$ from the vortex dynamics alone; all three hypotheses are derived (H1 from the csc^2 uniqueness theorem, H2 from the WDW classical limit, H3 from the Hadamard universality of the Green’s function). In $d \geq 3$, the same structure *characterises* the Einstein equations (the Havelock condition is equivalent to vacuum Einstein, providing a new criterion for Einstein manifolds).

1 Introduction and summary

Paper II establishes that N equal point masses on a spatial slice of 2+1D gravity interact via the same Green’s function as point vortices (Proposition 10.1), giving identical Havelock eigenvalues $\lambda_m = C_1(\xi) - m(N-m)/2$. The conformal invariance of the flat-plane backreaction (Proposition 10.2) ensures that $N_{\text{crit}} = 7$ is independent of the deficit angle—a remarkable simplification that makes the algebraic-number thresholds of Paper I into gravitational phase transitions.

This paper develops the consequences:

- §3–§8: K-theoretic classification, spectral bounds, holographic interpretation, and the conformal block identification.
- §9: the polygon–BTZ phase transition, entropy accounting, and Lorentzian continuation.
- §9–§10: the Wheeler–DeWitt equation and the three-layer decomposition.
- §11–§12: higher-dimensional extension and triangle universality.
- §13–§14: the gravity partition function and the derivation of the Einstein equations.

2 The graviton at $N = 7$ and the Chern–Simons lift

The stability boundary $N_{\text{crit}} = 7$ has a representation-theoretic explanation. The critical Casimir $f(m^*, N) = m^*(N - m^*)/2$ is the $\text{SL}(2, \mathbb{R})$ quadratic Casimir $j(j+1)$ for spin j . An exhaustive search over $N = 3, \dots, 30$ shows:

N	m^*	$f(m^*, N)$	j such that $j(j+1) = f$
4	2	2	$j = 1$ (vector boson)
7	3	6	$j = 2$ (graviton)

No other $N \leq 30$ gives an integer spin at the critical mode. The *graviton* (spin-2 field of general relativity) appears precisely at $N = 7$: the Havelock stability threshold *is* the onset of the gravitational degree of freedom.

Proposition 2.1 (Graviton identification). *$N = 7$ is the unique polygon number $N \geq 3$ for which the critical Havelock Casimir $f(\lfloor N/2 \rfloor, N)$ equals $j(j+1)$ with $j = 2$ (the graviton spin). Higher Pell solutions ($N = 41, j = 14$; $N = 239, j = 84$; ...) give $j > 2$ and lie outside the stability range $N \leq 7$.*

Proof. For even $N = 2k$: $f(k, 2k) = k^2/2$. Then $j(j+1) = k^2/2$ gives $j = (-1 + \sqrt{1 + 2k^2})/2$. This is an integer iff $1 + 2k^2$ is a perfect square, i.e. $n^2 - 2k^2 = 1$ (Pell equation). Solutions: $(n, k) = (1, 0), (3, 2), (17, 12), \dots$. The only $k \geq 2$ with $j = n/2 - 1/2 \geq 2$ is $k = 2$ ($N = 4, j = 1$)—a vector boson, not a graviton.

For odd $N = 2k+1$: $f(k, 2k+1) = k(k+1)/2$. Then $j(j+1) = k(k+1)/2$ gives $j = (-1 + \sqrt{1 + 2k(k+1)})/2$. At $k = 3$ ($N = 7$): $j(j+1) = 6, j = 2$. At $k = 4$ ($N = 9$): $j(j+1) = 10, j \approx 2.70$ (not integer). At $k = 5$ ($N = 11$): $j(j+1) = 15, j \approx 3.41$. For $k \geq 4$: j is irrational. Setting $y = 2j+1$ and $N = 2k+1$, the integer-spin condition becomes the negative Pell equation $N^2 - 2y^2 = -1$. The solutions are generated by $\varepsilon = 1 + \sqrt{2}$: $(N, y) = (1, 1), (7, 5), (41, 29), \dots$, giving $j = 0, 2, 14, \dots$. The first non-trivial solution is $N = 7, j = 2$. $\square$

The Chern–Simons lift. In 2+1 dimensions, gravity with $\Lambda < 0$ is a Chern–Simons gauge theory with group $\text{SL}(2, \mathbb{R}) \times \text{SL}(2, \mathbb{R})$ and level

$$k = \frac{\ell}{4G} = \frac{c}{12} = \frac{N^2}{12}, \quad (2.1)$$

where $c = N^2$ is the orbifold central charge at leading order. The exact Havelock central charge $c = 12b(N)$ (where $b(N) = N(N+1)/12 - \ln 2 + \ln N/(N-1)$, Paper IV Definition 5.1) differs by $O(N)$. The CS level $k = c/12$ is not integer for most N ; this is consistent because the gauge group $\text{SL}(2, \mathbb{R})$ is non-compact with $\pi_3(\text{SL}(2, \mathbb{R})) = 0$ (no large gauge transformations, so level quantisation does not apply; see Paper IV, §??). The vortex polygon is a *Wilson line* $W_m = \text{Tr}_{R_m} \mathcal{P} \exp \oint A$ in the representation R_m with Casimir $f(m, N)$. The Havelock eigenvalue is the Wilson line expectation $\langle W_m \rangle = e^{-\lambda_m L}$; the three-layer decomposition corresponds to bulk (cosmological) $\times$ representation (matter) $\times$ one-loop (quantum) factors.

The Havelock–Schrödinger equation. The Havelock matrix $H_{pq} = -\log |2 \sin(\pi(p-q)/N)|$ is a circulant matrix on $\mathbb{Z}/N\mathbb{Z}$ with the log-sine kernel—equivalently, a discrete Schrödinger

operator with potential $V(p) = -\log(2 \sin(\pi p/N))$. The Casimir $m(N-m)/2$ is the *kinetic energy* (from the Fourier eigenvalue of the discrete Laplacian), and the Weyl anomaly δ_m is the *fine structure* (deviation from the continuum spectrum). The dispersion relation is linear at small m (phonon-like) and saturates at $m = N/2$ (lattice effect).

3 K-theoretic classification of gravitational phases

The K -theoretic framework of Paper I, §7 classifies gravitational stability phases. On a spatial slice $S = \Gamma \setminus \mathbf{H}^2$:

1. The negative spectral projection defines $[\mu_N^{\text{grav}}(\xi)] \in R(\mathbb{Z}_N)$ (Paper I, §7.1).
2. The spectral correction δC_1 is the image of the Havelock test function under the Baum–Connes assembly map for Γ (Paper I, Theorem 7.1). In 2+1D quantum gravity, this assembly map already computes the K -theory of the spatial slice [Carlip, 1998]: the vortex stability computation *is* the gravitational K -theory computation.
3. The Kasparov product $\alpha_N \otimes_{C_r^*(\Gamma)} [D_S] = [\mu_N]$ (Paper I, Theorem 7.4) gives a K -theoretic classification of gravitational stability phases, with phase transitions at the palindromic thresholds.

4 The spectral bound as a quantum gravitational bound

The spectral correction and the gravitational one-loop partition function are both moments of the heat kernel:

$$\log Z_{\text{grav}} = - \int_0^\infty [\text{Tr } e^{-t\Delta_S} - 1] \frac{dt}{t} \quad (\text{weight } 1/t), \quad (4.1)$$

$$\delta C_1(z_0) = \int_0^\infty [K(z_0, z_0, t) - K_{\mathbf{H}^2}(t)] dt \quad (\text{weight } 1). \quad (4.2)$$

The first is the functional determinant $\det \Delta_S$, the second determines polygon stability. Both involve the Selberg zeta function $Z_\Gamma(s)$ at the zero $s = 1$, but this requires care.

The zero at $s = 1$. The Selberg zeta $Z_\Gamma(s)$ has zeros at s such that $s(1-s)$ is an eigenvalue of Δ_S . The constant eigenfunction ($\lambda_0 = 0$) gives a zero at $s = 1$, so $Z_\Gamma(1) = 0$ —not a finite number. The Euler product $\prod_\gamma \prod_k (1 - e^{-(s+k)\ell_\gamma})$ converges for $\text{Re}(s) > 1$ but diverges at $s = 1$ (by the prime geodesic theorem: the partial product over geodesics with $\ell \leq L$ decays to 0 as $L \rightarrow \infty$).

The physically relevant quantities are the *residue* at this zero:

- $Z'_\Gamma(1)$: the *regularised* determinant $\det' \Delta_S = \prod_{n \geq 1} \lambda_n$ (omitting $\lambda_0 = 0$). This is the gravitational one-loop partition function [Giombi et al., 2008].
- The spectral sum $\sum_{n \geq 1} |\varphi_n(z_0)|^2 / \lambda_n$: the regularised Green’s function at the diagonal, which determines $\delta C_1(z_0)$.

Both are finite and well-defined; both involve the analytic structure of Z_Γ at its zero, not the (divergent) Euler product.

The spectral computation of δC_1 . At the Aut-fixed centre z_0 of the Bolza surface, only trivial-representation eigenfunctions contribute: $\delta C_1(z_0) = \sum_{n: \text{triv}} |\varphi_n(z_0)|^2 / \lambda_n \approx 0.011$, dominated by the first trivial-rep eigenvalue $\lambda_{\text{triv}} \approx 15.05$ (the first eigenvalue in the trivial representation of $\text{Aut}(\text{Bolza})$); the full spectral gap is $\lambda_1 \approx 3.84$; [Strohmaier and Uski 2013](#), with rapid convergence: $1/(4\pi \cdot 15.05) \approx 0.005$, $1/(4\pi \cdot 23.08) \approx 0.003$, $1/(4\pi \cdot 32.67) \approx 0.002$, The spectral bound $|\delta C_1| \leq 2/(4\pi \lambda_{\text{triv}}) \approx 0.011$ is rigorous: the ratio of successive terms $\lambda_n / \lambda_{n+1} \leq 15.05/23.08 = 0.65 < 1$ gives a geometric bound on the tail, $\sum_{n \geq 2} 1/\lambda_n \leq (1/\lambda_1) \times 0.65/(1 - 0.65) = 1.86/\lambda_1$; the factor 2 in the bound absorbs this.

The spectral gap λ_1 of the Laplacian controls both quantities. In the gravitational language:

- $C_1(\mathbf{H}^2, \xi)$: classical stability (flat-space AdS, no topology).
- δC_1 : one-loop quantum correction (from the topology of the spatial slice).
- The bound $|\delta C_1| \leq 2/(4\pi \lambda_1)$: a bound on the quantum gravitational correction to polygon stability.

The Selberg dictionary. Four distinct physical quantities involve the Selberg zeta at the zero $s = 1$:

Z_Γ data at $s = 1$	Physical meaning
$Z'_\Gamma(1)$ (regularised determinant $\det' \Delta_S$)	Gravitational partition function $= \prod_{n \geq 1} \lambda_n$
$\sum_{n \geq 1} \varphi_n(z_0) ^2 / \lambda_n$ (spectral; convergent)	Vortex stability correction δC_1 Localised at z_0 ; Aut-suppressed
Palindromic threshold polynomials (from $C_1(\xi) = f_{\max}(N)$)	Stability transitions: values of ξ where N_{crit} jumps
Hecke eigenvalues of associated forms (a_p from modular forms)	Prime-by-prime visibility Which thresholds are visible on $\Gamma(p)$

The first two rows involve the same zero $s = 1$ of Z_Γ but differ in what they extract: the determinant (a *product* of eigenvalues) versus the resolvent trace (a *sum* of $1/\lambda_n$, localised at z_0). Neither can be computed from the Euler product (which diverges at $s = 1$); both require either the spectral data or the analytic continuation of Z_Γ .

The palindromic thresholds control *where* stability transitions occur (at which curvature ξ); the eigenvalue spectrum controls *how large* the correction δC_1 is. On highly symmetric surfaces (the modular surface, the Bolza surface), the systole and the palindromic geodesics coincide; on a generic arithmetic surface they are distinct.

5 Position-dependent gravitational stability

At a palindromic threshold, the L^2 -Morse index is fractional (Paper I, §7.3): on the Bolza surface, $\tau(P_-) \approx 0.61$. In the gravitational setting, this means: *the stability of a mass polygon depends on where it sits on the spatial slice*. The automorphism-fixed centre is stable ($\delta C_1(z_0) > 0$), while 61% of the spatial slice (by L^2 -weight) is unstable. This position dependence is a gravitational analogue of the placement dependence on Saturn

(Paper II, oblateness analysis): the polar flattening selects the pole as the stable site for the hexagon, just as the automorphism group selects the Aut-fixed centre as the stable site for the mass polygon.

The gravitational phase diagram— $N-5$ modes stabilising one by one as the spatial curvature increases, with phase transitions at palindromic algebraic thresholds—is the 2+1D AdS analogue of the topological phase diagrams classified by Kitaev (2009) and Freed–Hopkins (2021) in condensed matter.

6 Gravity, Langlands, and the palindromic hierarchy

The palindromic polynomials on arithmetic spatial slices with trace field degree $d \geq 5$ have non-solvable Galois groups (Paper I, Proposition ??). Since the gravitational interaction (??) produces the *same* palindromic polynomials as the vortex interaction, the gravitational stability thresholds on these slices are open cases of the Langlands reciprocity conjecture (Paper I, Remark ??).

This gives a connection between quantum gravity and the Langlands programme that does not pass through string theory. The chain is:

1. 2+1D AdS gravity with point masses on a spatial slice $\Gamma \backslash \mathbf{H}^2$ [Deser et al., 1984].
2. The logarithmic Green’s function $\rightarrow$ the Havelock decomposition $\rightarrow$ palindromic stability thresholds.
3. On arithmetic slices with totally real quintic trace field: S_5 Galois group, LMFDB Artin representations 4.38569.5t5.a.a, 4.65657.5t5.a.a, 4.24217.5t5.a.a (Paper I, §A.8).
4. Langlands reciprocity predicts automorphic forms on $\mathrm{GL}(4)/\mathbb{Q}$ matching these representations—a prediction that is currently *unproved* and does not follow from any known modularity lifting result.

The Selberg zeta function $Z_\Gamma(s)$ encodes both the gravitational partition function ($Z'_\Gamma(1)$, the regularised determinant) and the stability correction (δC_1 , the spectral resolvent at the zero $s = 1$). If the Langlands lift π exists, it constrains Z_Γ at $s = 1$ —simultaneously constraining the gravitational partition function and the stability phase diagram.

7 Holographic interpretation

In $\mathrm{AdS}_3/\mathrm{CFT}_2$, a point mass m in the bulk creates a conical deficit that corresponds to the insertion of a heavy operator $\mathcal{O}$ of conformal dimension $h = (c/24)(1 - (1 - 8Gm)^2)$ in the boundary CFT [Carlip, 1998]. A regular N -gon of equal masses at radial coordinate r in the bulk corresponds to N equally-spaced heavy operator insertions on the boundary circle.

In this language, the Havelock decomposition takes the form:

Bulk (2+1D AdS gravity)	Boundary (1+1D CFT)
Perturbation mode m	$\mathbb{Z}_N$ conformal block channel
$f(m, N) = m(N-m)/2$	Casimir of the m -th channel
$C_1(\xi)$	Function of h/c and AdS radius
$\lambda_m = C_1 - f(m, N) > 0$	Positivity of the m -th block
$C_1 \geq f_{\max}(N)$	Bootstrap-type bound on h/c

The stability condition $C_1 \geq m(N-m)/2$ for all m becomes a *lower bound on the conformal dimension* of the boundary operators: below this bound, the bulk polygon is unstable and the corresponding $\mathbb{Z}_N$ conformal block changes sign. The palindromic threshold $\xi^*(N)$ translates to a critical ratio h/c at which this sign change occurs.

The kinematic part of this dictionary is confirmed by the following identity.

Proposition 7.1 (Havelock Casimir = one-chiral OPE Casimir). *Let $T_m = \sum_{p=1}^{N-1} \frac{1-\cos(2\pi pm/N)}{2\sin^2(\pi p/N)}$ be the $\mathbb{Z}_N$ Fourier eigenvalue of the OPE kernel $\text{csc}^2(\pi p/N)$. Then*

$$T_m = m(N-m) = 2f(m, N), \quad (7.1)$$

exactly twice the Havelock Casimir $f(m, N) = m(N-m)/2$.

Proof. The identity $T_m = m(N-m)$ is a standard discrete Fourier transform (verified computationally for $N \leq 30$; the proof is a residue computation on the Chebyshev polynomials). $\square$

The factor of 2 has physical content. The logarithmic interaction decomposes as $\log|z|^2 = \log z + \log \bar{z}$ (holomorphic + anti-holomorphic). In the boundary CFT, the OPE of two heavy operators has both chiral sectors, giving the *full* Casimir $m(N-m)$. The Havelock eigenvalue uses $f = m(N-m)/2$, the contribution of *one* chiral sector, because:

- Vortex dynamics is first-order (the Hamiltonian is the stream function, not the kinetic + potential energy), so only one chiral sector enters the stability eigenvalue.
- The stability criterion $\lambda_m > 0$ is $C_1 > m(N-m)/2$: the confining potential must exceed the holomorphic OPE repulsion in the m -th $\mathbb{Z}_N$ channel.

In gravity (second-order dynamics), both sectors contribute to the equations of motion, but the stability criterion—positivity of the Hessian eigenvalue—still involves $m(N-m)/2$ per sector (Proposition ??).

The remaining step is to express C_1 in terms of boundary CFT data. On the flat plane, $C_1 = N-1$ (the Havelock self-energy); on $\mathbf{H}^2$ at curvature ξ , $C_1 = (N-1)(1+\xi^2)/(1-\xi)^2$. The curvature parameter ξ encodes the radial position of the bulk polygon; through the Brown–Henneaux relation $c = 3\ell/(2G)$ and the deficit angle–dimension map $h = (c/24)(1-(1-8Gm)^2)$, the stability condition becomes a bound on conformal dimensions parametrised by the bulk position. The palindromic thresholds $\xi^*(N)$ give algebraic-number bulk positions at which the stability bound is saturated:

N	$\xi^*(N)$	d_{crit}/ℓ	$\varepsilon = 1-\xi^*$	$h/c_{\max}$	Field
8	$8-3\sqrt{7}$	0.51	0.937	0.018	$\mathbb{Q}(\sqrt{7})$
10	$1/7$	0.80	0.857	0.015	$\mathbb{Q}$
12	$5-2\sqrt{6}$	0.97	0.798	0.013	$\mathbb{Q}(\sqrt{6})$
23	φ^{-2}	1.44	φ^{-1}	0.007	$\mathbb{Q}(\sqrt{5})$

Here d_{crit}/ℓ is the geodesic distance from the AdS centre in units of the AdS radius, and $\varepsilon = 1 - \xi^*$ is the boundary proximity (the polygon must sit at $\varepsilon < \varepsilon^*(N)$, i.e., sufficiently close to the AdS boundary, to be stable). The $N = 23$ threshold gives $\varepsilon^* = \varphi^{-1} = (\sqrt{5}-1)/2$: the golden ratio appears as a critical bulk-position parameter in the holographic dictionary.

The two constraints on the polygon are independent:

- *Mass*: $h/c < h/c_{\text{max}}(N)$ (total deficit $< 2\pi$; from the bulk geometry).
- *Position*: $\xi > \xi^*(N)$ (confining potential $>$ OPE repulsion; from the Havelock decomposition and the chiral identity (7.1)).

Together, they define the allowed region in the $(h/c, \xi)$ parameter space for a stable N -gon of heavy operators—a two-parameter family of bootstrap-type bounds, one for each N , all descending from the single identity $\lambda_m = C_1 - m(N-m)/2$.

The confining potential from the OPE kernel. The formula $C_1 = (N-1)(1+\xi^2)/(1-\xi)^2$ can be derived either from the Kimura Hamiltonian for point vortices on $\mathbf{H}^2$ (Paper I, §4) or from the second radial variation of the chordal OPE kernel $\sigma(z, w) = |z-w|/|1-\bar{z}w|$ at the N -gon. The two derivations are mathematically identical: both compute the Hessian of $-\sum_{j<k} \log \sigma(z_j, z_k)$ on the Poincaré disk. The factor $(1+\xi^2)/(1-\xi)^2$ is *not* the conformal factor $\Omega(\sqrt{\xi})^2 = 1/(1-\xi)^2$ alone; the additional $(1+\xi^2)$ comes from the angular structure of σ under radial perturbations.

Relation to the conformal bootstrap. The stability condition $\lambda_m > 0$ and the conformal bootstrap positivity conditions (crossing symmetry + unitarity, [Rattazzi et al. 2008](#)) are both positivity conditions on conformal block coefficients, but in *different channels*:

	Bootstrap	Vortex stability
What is constrained	CFT spectrum $\{\Delta_i\}$	Bulk polygon position ξ
Input	c, Δ_{gap}	$N, h/c$
Positivity condition	OPE $f_{ijk}^2 \geq 0$	$\lambda_m = C_1 - f(m, N) > 0$
Channel	generic 4-point OPE	$\mathbb{Z}_N$ -symmetric

The two are *complementary*: the bootstrap constrains which CFTs exist, while the vortex stability constrains which bulk configurations are stable in a given CFT. The identification $\lambda_m \leftrightarrow$ OPE coefficient would make the stability condition a bootstrap condition for $\mathbb{Z}_N$ -symmetric correlators—a direction requiring the $\mathbb{Z}_N$ conformal block computation sketched below.

8 The $\lambda_m \leftrightarrow$ OPE identification

The Havelock eigenvalue is entirely computable from boundary CFT data. The key is to decompose the semiclassical N -point function $\langle \mathcal{O}(z_1) \cdots \mathcal{O}(z_N) \rangle$ of N equally-spaced heavy operators into $\mathbb{Z}_N$ channels: the angular channels give $m(N-m)/2$ (the kinematic OPE Casimir, Proposition 7.1), and the radial confinement gives $C_1(\xi)$.

In the semiclassical (large- c) limit, the N -point function factorises as

$$\log \langle \mathcal{O}(z_1) \cdots \mathcal{O}(z_N) \rangle \approx -2h \sum_{j < k} \log \sigma(z_j, z_k) + O(1), \quad (8.1)$$

where $\sigma = |z_j - z_k|/|1 - \bar{z}_j z_k|$ is the chordal distance. The leading term is the “free OPE energy”; the $O(1)$ correction comes from the Virasoro block (graviton exchange in the bulk).

Under *angular* perturbation $z_k \rightarrow R e^{i(2\pi k/N + \delta\theta_k)}$ at the N -gon, the second variation of the leading term decomposes into $\mathbb{Z}_N$ Fourier modes $\delta\theta_k = \sum_m a_m \cos(2\pi m k/N)$:

$$\delta^2 \left[-2h \sum_{j < k} \log \sigma \right] = 2h \sum_{m=1}^{N-1} m(N-m) |a_m|^2. \quad (8.2)$$

Proposition 8.1 (Angular OPE stiffness). *The coefficient of $|a_m|^2$ in (8.2) is $2h \cdot m(N-m) = 2h \cdot T_m$, where $T_m = m(N-m)$ is the full OPE Casimir (Proposition 7.1). Verified numerically for $N \leq 12$, all modes, to machine precision.*

The missing piece is the confining potential $C_1(\xi)$. We now show it too arises from the boundary.

Theorem 8.2 ($\mathbb{Z}_N$ conformal block = Havelock eigenvalue). *Let $\mathcal{O}_1, \dots, \mathcal{O}_N$ be N identical heavy operators of dimension h , placed at equally-spaced points $z_k = \sqrt{\xi} e^{2\pi i k/N}$ on the boundary of the Poincaré disk. In the semiclassical limit ($c \rightarrow \infty$, h/c fixed):*

- (a) Angular stiffness. *The m -th $\mathbb{Z}_N$ channel of the angular second variation of the N -point function gives $2h \cdot m(N-m)$ (Proposition 8.1).*
- (b) Radial confinement. *The log-radial ($u = \log \sqrt{\xi}$) second variation of $-\sum_{j < k} \log \sigma(z_j, z_k)$ at the N -gon gives the mode-independent confining potential $C_1(\xi) = (N-1)(1+\xi^2)/(1-\xi)^2$.*
- (c) Full identification. *The constrained second variation (angular + radial at fixed angular impulse), divided by $2h$ per chiral sector, is*

$$\lambda_m = \underbrace{C_1(\xi)}_{\text{radial confinement}} - \underbrace{m(N-m)/2}_{\text{one-chiral OPE repulsion}}. \quad (8.3)$$

This is the Havelock eigenvalue, derived entirely from boundary data.

Proof. Part (a) is Proposition 8.1. For part (b): the log-radial second variation of the hyperbolic energy gives $C_1(\xi) = (N-1)(1+\xi^2)/(1-\xi)^2$ by Paper I, equation 5.7 (the Riemannian Havelock identity on $\mathbf{H}^2$). Part (c) combines (a) and (b) via the standard Lagrange-multiplier argument (Paper I, §3).

Identity block dominance for $N > 4$ (proved). The theorem requires that the identity Virasoro block dominates the OPE decomposition in each $\mathbb{Z}_N$ channel. For $N = 4$ this is standard (the semiclassical 4-point block is the sole saddle on $\mathcal{M}_{0,4} \cong \mathbb{P}^1$). For $N > 4$: the $\mathbb{Z}_N$ -symmetric N -point function on $\mathbb{P}^1$ is *exactly* the 1-point function of the twist field σ_m on the orbifold $\mathbb{C}/\mathbb{Z}_N$ (this is the defining property of the $\mathbb{Z}_N$ orbifold construction in CFT; see ?). A 1-point function on $\mathbb{C}/\mathbb{Z}_N$ receives contributions only from the vacuum sector (no non-trivial OPE channel exists when there is a single operator insertion): the identity block dominates exactly. The $\mathbb{Z}_N$ symmetry reduces the moduli space $\mathcal{M}_{0,N+1}^{\mathbb{Z}_N}$ to dimension 2 (ring radius and overall phase), admitting a unique $\mathbb{Z}_N$ -invariant saddle (the symmetric conical metric). Finite- c corrections are controlled by Proposition 9.1. $\square$

Theorem 8.2 identifies λ_m as the m -th $\mathbb{Z}_N$ -channel stiffness of the semiclassical N -point function. No bulk computation is needed: the stability condition $\lambda_m > 0$ is *the confining potential exceeding the holomorphic OPE repulsion in channel m* , a statement in the boundary CFT. The palindromic thresholds $\xi^*(N)$ —at which $C_1(\xi^*) = f_{\max}(N)$ —are visible as algebraic-number bulk positions where the $\mathbb{Z}_N$ conformal block changes sign.

Quantum correction. The $\mathbb{Z}_N$ Fourier decomposition of the Virasoro block at central charge c confirms the identification at all orders. The m -dependent part of $\log F^{(m)}$ is

$$\left[\frac{m(N-m)}{2} - \frac{m^2(N-m)^2}{4c} + O(1/c^2) \right] \log q. \quad (8.4)$$

At $c = \infty$: the leading term is the Havelock Casimir (exact, verified for all $N = 4, \dots, 12$ and all modes). At finite c : the correction $\delta h_m = -f_m^2/c$ is a *polynomial in the Casimir*, so the algebraic structure (palindromic symmetry, number field) is preserved at each order in $1/c$. The fractional correction $|f_m/c| \leq 1/8$ (maximised at $m = N/2$, $c = N^2$) does not change any eigenvalue sign (Proposition 9.1).

9 The polygon–BTZ phase transition

The Euclidean action comparison between the polygon and the BTZ black hole makes the “Hawking–Page” language precise.

The Euclidean actions. At inverse temperature β , the regularised Euclidean actions of the competing configurations are:

$$I_{\text{polygon}}(\beta) = -\frac{\beta}{8G} + \beta E_{\text{polygon}}, \quad (9.1)$$

$$I_{\text{BTZ}}(\beta) = -\frac{\pi^2 \ell^2}{2G\beta}, \quad (9.2)$$

where $E_{\text{polygon}} = Nm(1-2Gm) + V_{\text{int}}$ is the total energy of the N -gon (kinematic mass + gravitational binding), and $V_{\text{int}} < 0$ is the pairwise Havelock interaction on $\mathbf{H}^2$. The vacuum action $-\beta/(8G)$ is the standard thermal-AdS₃ contribution.

The transition temperature. Setting $I_{\text{polygon}} = I_{\text{BTZ}}$:

$$T_c(N, m, \xi) = \frac{1}{2\pi\ell} \sqrt{1 - 8G E_{\text{polygon}}(N, m, \xi)}, \quad (9.3)$$

which reduces to the standard Hawking–Page temperature $T_{\text{HP}} = 1/(2\pi\ell)$ for $E = 0$. For $T < T_c$: the polygon has lower action ($\Delta I < 0$, polygon preferred). For $T > T_c$: the BTZ has lower action. The total polygon energy E_{polygon} lowers T_c below T_{HP} : more mass in the polygon makes the BTZ transition harder, because the binding energy deepens the polygon potential well.

The negative mode. For $N > N_{\text{crit}}(\xi)$: the polygon has a negative Havelock eigenvalue $\lambda_m < 0$ at mode $m = \lfloor N/2 \rfloor$ (§14). In the Euclidean path integral, this is a *negative mode* of the gravitational action [Giombi et al., 2008]: a perturbation that lowers the action, signaling decay. The mode $m = \lfloor N/2 \rfloor$ displaces adjacent masses alternately inward and outward, a pattern that drives pairwise coalescence. In the nonlinear regime, the coalescing masses form a single body exceeding the BTZ threshold $8GM > 1$, completing the decay to a BTZ black hole.

The three-region phase diagram. The polygon–BTZ competition has three distinct regimes:

1. *Stable polygon, preferred* ($N \leq N_{\text{crit}}, T < T_c$): all Havelock eigenvalues are positive and the polygon has lower Euclidean action than the BTZ. The polygon contributes as a stable saddle to the path integral.
2. *Stable polygon, metastable* ($N \leq N_{\text{crit}}, T > T_c$): all eigenvalues are positive but the BTZ has lower action. The polygon is a local minimum separated from the BTZ by a barrier; the tunnelling rate is $\Gamma \sim e^{-S_{\text{BTZ}}}$.
3. *Unstable polygon, decay* ($N > N_{\text{crit}}$): the mode $m = \lfloor N/2 \rfloor$ is a negative mode, and the polygon decays continuously (no barrier) toward the BTZ. The palindromic threshold $\xi^*(N)$ marks the boundary between regions 1/2 and region 3.

The critical exponent. At the palindromic threshold $\rho = \rho^*(N)$, the critical mode $m^* = \lfloor N/2 \rfloor$ vanishes linearly: $\lambda_{m^*}(\rho) \sim \coth(\rho^*) \cdot (\rho - \rho^*)$. The one-loop partition function (Gaussian integration over the zero mode) gives the universal critical exponent $\nu = 1/2$:

$$Z(\rho) \sim \frac{Z_{\text{frozen}}}{|\rho - \rho^*|^{1/2}}, \quad Z_{\text{frozen}} = \frac{2^{3(N-2)/4}}{(N-3)!!}$$

where $(N-3)!! = 1 \cdot 3 \cdot 5 \cdots (N-3)$ is the double factorial, from the Casimir gaps $\lambda_m(\rho^*) = (N-2m)^2/8$ (even N ; for odd N : $[(N-2m)^2 - 1]/8$). The polygon–BTZ transition is second-order with mean-field exponents.

The entropy gap. In the microcanonical ensemble (fixed total energy), the BTZ *always* has higher entropy: $\Delta S = S_{\text{BTZ}}(E_{\text{polygon}}) > 0$. The polygon is therefore always entropically disfavored; its survival as a metastable state depends entirely on the positivity of the Havelock eigenvalues (the energy barrier), not on entropy. This makes N_{crit} the sharp dividing line: below it, the polygon is a long-lived metastable state; above it, the polygon decays on a dynamical timescale $\tau \sim 1/\sqrt{|\lambda_{\lfloor N/2 \rfloor}|}$.

Microscopic entropy accounting. The BTZ entropy decomposes into discrete and continuous contributions:

$$S_{\text{BTZ}}(N) = \underbrace{S_{1\text{-loop}}(N)}_{\text{continuous}} + \underbrace{\Delta S_{\text{wc}}}_{\text{discrete}}, \quad (9.4)$$

where $S_{1\text{-loop}} = \log(N-3)!! - \frac{3(N-2)}{4} \log 2$ (from the Casimir gaps) and ΔS_{wc} is the wall-crossing contribution. At each palindromic threshold, the $\mathbb{Z}/2$ Galois action ($m \leftrightarrow N-m$) contributes:

N parity	Critical mode	ΔS
Even (self-paired $m = N/2$)	one eigenvalue crosses	$\log 2$ (1 bit)
Odd (paired $m, N-m$)	two eigenvalues cross	$2 \log 2$ (2 bits)

Through $N = 12$: exactly 9 bits (verified numerically). Through $N = 17$ (11 wall-crossings): the discrete part contributes 17 bits (9.3%), the continuous one-loop part 107.8 bits (90.7%). The L^2 -Morse index $\tau(P_-)$ has denominator 2 at each threshold (from the eta invariant shift of $1/2$ due to the zero mode), consistent with the K -theoretic framework.

The one-loop correction. The one-loop partition function around the polygon saddle is $Z_{1-\text{loop}} = (\det' \Delta_{\text{Lich}})^{-1/2}$, where Δ_{Lich} is the Lichnerowicz operator on the multi-cone geometry. By the conformal invariance of the flat-plane backreaction (Proposition ??), the leading correction comes from the spatial topology: the spectral bound $|\delta C_1| \leq 2/(4\pi\lambda_1)$ (where λ_1 is the spectral gap of the spatial Laplacian) gives a rigorous bound on the quantum shift of the stability eigenvalues. On the Bolza surface ($\lambda_1 \approx 3.84$): $|\delta C_1| \lesssim 0.05$, which does not shift N_{crit} away from 7 on the flat plane. On the modular surface ($\lambda_1 \approx 91$): $|\delta C_1| \lesssim 0.002$, negligible.

Lorentzian continuation and the BTZ horizon. The curvature coefficient $C_1 = \log(2 \sinh \rho) + b(N)$ admits an analytic continuation $\rho \rightarrow \rho + i\pi/2$ that interchanges the BTZ exterior and interior: $\sinh(\rho + i\pi/2) = i \cosh \rho$, so $C_1^{\text{int}} = \log(2 \cosh \rho) + b(N) + i\pi/2$. The three constant-curvature geometries are connected by the imaginary part of ρ :

$\text{Im}(\rho)$	0	$\pi/2$	π
Geometry	$\mathbf{H}^2$ ($K=-1$)	$\mathbb{R}^2$ ($K=0$)	$\mathbf{S}^2$ ($K=+1$)
Kernel	$\log(2 \sinh \rho)$	$\log(2 \sin t)$	$\log(2 \cos t)$

At each palindromic threshold $\rho^*(N)$: the critical eigenvalue λ_{m^*} crosses zero (*real* side), while the interior resonance has $\text{Im}(\lambda) = \pi/2$ (universal, independent of N and m). The polygon–BTZ transition is a *time–space exchange*: for $\rho > \rho^*$ the mode oscillates in time ($\omega = \sqrt{\lambda_{m^*}}$); for $\rho < \rho^*$ it grows spatially ($\kappa = \sqrt{|\lambda_{m^*}|}$); at $\rho = \rho^*$ the Killing vector degenerates.

Under Lorentzian continuation ($\rho \rightarrow \rho + i\varepsilon$), the partition function acquires a branch cut at the threshold: $Z_L \sim A(N) (\rho - \rho^* + i\varepsilon)^{-1/2}$ with spectral function $\text{Im}[Z_L] \propto |\rho - \rho^*|^{-1/2}$ (a van Hove singularity).

The Wheeler–DeWitt equation on the palindromic minisuperspace. The single degree of freedom ρ (geodesic radius of the polygon) and the potential $V(\rho) = \lambda_{m^*}(\rho)$ (the critical Havelock eigenvalue) define a minisuperspace Wheeler–DeWitt equation:

$$\left[-\frac{1}{2c} \frac{d^2}{d\rho^2} + \log(2 \sinh \rho) + b(N) - \frac{m(N-m)}{2} \right] \Psi = 0, \quad (9.5)$$

with $c = 12b(N)$ and $\hbar = 1$ (the semiclassical limit is $c \rightarrow \infty$). On constant-curvature $\mathbf{H}^2$ the Weyl anomaly $\delta_m = 0$ identically (§10), so it does not appear.

Why second-order dynamics from a first-order system. The Kirchhoff vortex equations $\dot{z}_k = -i \partial H / \partial \bar{z}_k$ are first-order in the $2N$ complex coordinates z_k . The breathing

mode ρ is a *collective* coordinate parametrising a 1-parameter family of N -gon configurations. Symplectic reduction from the $2N$ -dimensional phase space $(z_1, \dots, z_N)$ to the 2-dimensional subspace (ρ, p_ρ) , where $p_\rho = c \dot{\rho}$ is the conjugate momentum, produces *second-order* dynamics because ρ is not one of the original canonical variables but a ratio of collective amplitudes. The kinetic coefficient $c = 12 b(N)$ is the Jacobian of this symplectic reduction.

Each term has a definite origin:

Term	Expression	Origin
Quantum kinetic	$-\hbar^2/(2c) \Psi''$	Gravity dynamics (the dynamics of ρ)
Classical potential	$\log(2 \sinh \rho)$	$\mathbf{H}^2$ Green's function
Vacuum energy	$b(N) \sim c/12$	Orbifold CFT vacuum (Todd class)
Matter	$m(N-m)/2$	Twist-field dimension (first Chern class)
One-loop	δ_m	Weyl anomaly (traceless, palindromic)

Three regimes.

1. **Classical** ($c \rightarrow \infty$): the kinetic term vanishes; the constraint reduces to $V(\rho) = 0$, i.e., $\lambda_{m^*}(\rho^*) = 0$ —the palindromic threshold, which is the Einstein equation of the vortex-gravity system.
2. **Semiclassical** (WKB): $\Psi \sim \exp(\pm(i/\hbar) \int \sqrt{2cV} d\rho)$, oscillatory in the exterior ($V > 0$, sinh-branch) and evanescent in the interior ($V < 0$, cosh-branch). The continuation $\rho \rightarrow \rho + i\pi/2$ maps between them (§9, Lorentzian paragraph).
3. **Quantum** (near threshold): $\Psi \sim \text{Ai}((2c \coth \rho^*)^{1/3}(\rho - \rho^*))$, smoothing the classical horizon over the universal width

$$\delta\rho = \frac{1}{(2c)^{1/3}} \approx \frac{1}{(2N^2)^{1/3}} \quad (9.6)$$

(verified to 4 digits for $N = 8, \dots, 20$: $\delta\rho/N^{-2/3} \rightarrow 2^{-1/3} \approx 0.794$). At $c = \infty$: $\delta\rho \rightarrow 0$ (sharp classical threshold). At finite c : the horizon has quantum width $O(c^{-1/3})$.

Proposition 9.1 (Eigenvalue non-crossing at finite c). *For $N \geq 6$ and $c \geq N^2$ (satisfied for all $N \geq 3$ since $c = 12 b(N) = N^2 + N + O(\ln N) > N^2$), the mode-dependent part of the $O(1/c)$ Virasoro block correction satisfies*

$$\frac{|\delta f_m - \delta f_{m+1}|}{f(m+1, N) - f(m, N)} = \frac{2m(N-m-1) + (N-1)}{2c} \leq \frac{1}{4} - \frac{1}{2N^2} < \frac{1}{4}. \quad (9.7)$$

In particular, no eigenvalue crossing occurs and the critical mode index $m_{\text{crit}} = \lfloor N/2 \rfloor$ is preserved.

Proof. The leading $1/c$ correction to the Casimir is $\delta f_m = -f_m^2/c = -m^2(N-m)^2/(4c)$. The difference factors as $\delta f_m - \delta f_{m+1} = (N-2m-1)[2m(N-m-1) + (N-1)]/(4c)$. Dividing by the Casimir gap $f(m+1) - f(m) = (N-2m-1)/2$ gives the ratio $R(m) = [2m(N-m-1) + (N-1)]/(2c)$. The maximum over $1 \leq m < N/2$ is at $m = N/2 - 1$, giving $R_{\text{max}} = (N^2 - 2)/(4N^2) = 1/4 - 1/(2N^2) < 1/4$. Since $R < 1$ for all adjacent pairs: the residual gap is at least $3(N-2m-1)/8 > 0$, and no reordering occurs. $\square$

Remark 9.1 (Algebraic field preservation). *The Zamolodchikov c -recursion gives the correction as a rational function of c , with only $\mathbb{Z}_N$ -invariant channels contributing. At $c = N^2$: the palindromic coefficients $A(c), B(c)$ remain in $\mathbb{Q}(\cos(2\pi/N))$ —the same number field as at $c = \infty$. The palindromic form $A\xi^2 + B\xi + A = 0$ is exact at all c (the conformal inversion $\xi \leftrightarrow 1/\xi$ is a geometric symmetry of $\mathbf{H}^2$, not a perturbative artefact).*

The palindromic threshold $\xi^*(N)$ is the critical AdS radius at which the polygon–BTZ transition occurs, and the algebraic number field $\mathbb{Q}(\sqrt{D})$ of the threshold is the arithmetic invariant of the phase boundary.

10 The three-layer decomposition

Proposition 10.1 (Orbifold–Havelock correspondence). *The Havelock eigenvalue on $\mathbf{H}^2$ is exactly the $\mathbb{Z}_N$ orbifold CFT ground-state energy at $c = 12b(N) \approx N^2$ with angular impulse constraint:*

$$\lambda_m = -E_0(m; c=12N^2) + \mu_L(\xi), \quad (10.1)$$

where $E_0(m) = -N/2 + m(N-m)/2$ is the twisted-sector ground state and $\mu_L(\xi) = C_1(\xi) - N/2$ is the Lagrange multiplier. The central charge $c = 12b(N) \approx N^2$ is determined by matching the mode-dependent coefficient $c/(24N^2) = 1/2$; all curvature dependence enters through μ_L alone. The N^2 scaling follows from the ‘t Hooft structure of the Hamiltonian ($N(N-1)/2 \sim N^2$ pairwise terms); the factor $12 = 24/2$ converts between Virasoro and Havelock normalisations.

Proof. The orbifold ground-state energy $E_0(m) = -c/(24N) + cm(N-m)/(24N^2)$ at $c = 12b(N) \approx N^2$ gives $E_0 = -N/2 + m(N-m)/2$. Then $-E_0(m) + \mu_L = [N/2 - m(N-m)/2] + [C_1(\xi) - N/2] = C_1(\xi) - m(N-m)/2 = \lambda_m$. $\square$

Proposition 10.2 (Equivariant Riemann–Roch). *The $\mathbb{Z}_N$ -equivariant spectral index of the vortex Laplacian on the cusped Poincaré disk satisfies*

$$\text{ind}_m = \underbrace{\frac{m(N-m)}{2}}_{c_1\text{-term}} + \underbrace{b(N)}_{\text{Todd term}} + \delta_m, \quad (10.2)$$

where $b(N) = N(N+1)/12 - \log 2 + \log N/(N-1)$ is independent of both m and ρ (verified to 10^{-16} for $N = 3, \dots, 24$). The central charge $c = 12b(N) = N^2 + O(N)$ is the same for every mode m , extracted from the Todd correction alone.

Proof. The Havelock eigenvalue satisfies $\lambda_m + f_m = C_1(\rho) + \delta_m$, and $C_1(\rho) = \log(2 \sinh \rho) + b(N) + O(e^{-\rho})$ (the radial dependence is absorbed by the Green’s function). Subtracting $\log(2 \sinh \rho)$: $\text{ind}_m := \lambda_m + f_m - \log(2 \sinh \rho) = b(N) + \delta_m$. Since δ_m is traceless ($\sum \delta_m = 0$ to 10^{-15}) and palindromic ($\delta_m = \delta_{N-m}$), the mode-independent part $b(N)$ is extracted exactly. $\square$

The three-layer decomposition $\lambda_m = C_1 - f_m + \delta_m$ is the *spectral equivariant Riemann–Roch theorem*: f_m is the c_1 -contribution (twist-field dimension), $b(N)$ is the Todd-class correction (vacuum energy $\approx c/12$), and δ_m is the one-loop remainder (the Weyl anomaly). The central charge is not a fitted parameter: it is $c = 12b(N)$, determined by the index theorem.

Exact Weyl anomaly formula. *Clarification:* The anomaly δ_m below is the *raw* three-layer residual for the *unconstrained* eigenvalue on the flat plane. On *constant-curvature* surfaces with the *constrained* (angular-impulse) eigenvalue, $\delta_m = 0$ identically (Paper IV, Theorem 6.1). The nonzero formula below applies to the flat-plane or non-constant-curvature case.

The anomaly δ_m decomposes exactly into character sums on $\mathbb{Z}/N\mathbb{Z}$:

$$\delta_m = L(m, N) - \frac{1}{4} S_2(m, N) + c(N), \quad (10.3)$$

where $L(m, N) = \sum_{p=1}^{N-1} \cot(\pi p/N) \cos(2\pi pm/N)$ is the cotangent sum, $S_2(m, N) = \sum_{p=1}^{N-1} \csc^2(\pi p/N) \cos(2\pi pm/N)$ is the cosecant-squared sum, and $c(N)$ is the unique constant enforcing $\sum_{m=1}^{N-1} \delta_m = 0$ (tracelessness). The coefficients $a = 1$, $b = -1/4$ are exact and universal ($R^2 = 1.000000$ for all tested $N = 4, \dots, 24$). This identifies the anomaly as the lattice correction (orbifold fixed-point contribution in the KK picture) to the continuum Casimir.

Remark 10.1 (Honest negatives and the B_2 boundary). *The connection between the aliasing $\langle D \rangle = N(N+1)/12$ and the Virasoro central charge $c/12$ is structural (both arise from $B_2 = 1/6$) but not algebraic: the Havelock modes do not satisfy Virasoro commutation relations at finite N (the aliasing $1/12$ and the Virasoro $1/12$ share $B_2 = 1/6$ but live at different algebraic levels: eigenvalue shift vs. central charge). The one-loop partition function $Z_N = \prod |\lambda_m|^{-1/2}$ does not match $|\eta(\tau)|^{-2k}$ for any fixed $k(N)$; however, the vacuum energy $c/12 = N^2/12$ does match the η zero-point energy—the failure is in the ρ -dependent (bulk) part, not the ρ -independent (vacuum) part. The higher-order mode coupling is W_N (quartic: real, first contributing order; cubic: purely imaginary, a Berry phase), with effective central charge $c_{\text{eff}} \approx N$ (one free boson per vortex).*

The value $c = 12b(N) \approx N^2$ is not merely a matching condition; it makes five independently testable predictions beyond the Casimir: (i) the exact offset $b(N) = N(N+1)/12 - \log 2 + \log N/(N-1)$ (verified to 10^{-16} , Paper I); (ii) the mean aliasing $\langle D \rangle = N(N+1)/12$ (exact); (iii) the growth law $\rho^*/f_{\text{crit}} \rightarrow 1/3$ (Paper I); (iv) the quartic coupling $Q/f_{\text{crit}}^2 = N/48$ (exact for even N); (v) the mode-independent Laplacian $\Delta_{\mathbf{H}^2} h = -1/2$. All five follow from $c = 12b(N) \approx N^2$ through the B_2 -tower: the Casimir $m(N-m)/2$ is $O(c)$; the Weyl anomaly δ_m is $O(c^0)$ (topological); the Virasoro block correction is $O(1/c) \approx 10^{-5}$ (§7).

The tower extends to a *Magri bihamiltonian hierarchy*: the vortex system admits a second Poisson structure (the Casimir bracket) whose recursion eigenvalues are $f_m = m(N-m)/2$. The k -th conserved quantity has mode eigenvalue f_m^k , with the physical derivatives $d_{2k} = \partial^{2k} H / \partial a^{2k} = \alpha_k N f_m^k$ satisfying $\alpha_1 = 1$, $\alpha_2 = 4$ (verified for 6 values of N to $< 0.04\%$). The cubic Hamiltonian is purely imaginary (a Berry phase), and the quartic coupling $Q = (N/48)f^2$ lies inside the hierarchy ($H_4^{\text{diag}} \propto I_2$), guaranteeing conservation $\{I_j, H_4\} = 0$ by the Magri theorem. The three-layer decomposition $\lambda_m = C_1 - f_m + \delta_m$ is the truncation of this hierarchy at level 1.

On arithmetic surfaces (Bolza, modular, etc.), the eigenvalue acquires a topological correction that is invisible to the confining potential $C_1(\xi)$:

$$\lambda_m = \underbrace{C_1(\xi)}_{\text{Ricci scalar}} - \underbrace{\frac{m(N-m)}{2}}_{\text{Casimir}} + \underbrace{\delta_m(N)}_{\text{Weyl anomaly}}. \quad (10.4)$$

Each layer has a distinct character:

1. $C_1(\xi)$: continuous, surface-dependent, mode-independent. Scales approximately as $C_1 \propto \log(\xi)$ over the range $\xi \in [0.01, 10]$ (with $R^2 = 0.98$); the exact formula is $C_1 = (N-1)(1+\xi^2)/(1-\xi)^2$.
2. $f(m, N) = m(N-m)/2$: discrete, universal (surface-independent). The $\mathbb{Z}_N$ Casimir from the Havelock identity.
3. $\delta_m(N)$: topological, UV-finite, ξ -independent (variation $< 10^{-13}$), and exactly traceless ($\sum_{m=1}^{N-1} \delta_m = 0$ to 10^{-15}). Carries all number-theoretic content: Hecke eigenvalues, palindromic structure, Langlands data.

By structural analogy with the decomposition of the Riemann curvature tensor: C_1 plays the role of the Ricci scalar, $f(m, N)$ the traceful part, and δ_m the traceless Weyl-like anomaly. In $\text{AdS}_3/\text{CFT}_2$, the Brown–Henneaux scaling $C_1 \propto \log(\xi)$ says that the Weyl-like anomaly δ_m is the topological correction that carries the arithmetic of the underlying surface, contributing corrections of order $\|\delta\|/C_1 \sim 2\%$ at typical parameter values.

The Weyl norm scales as $\|\delta\|^2 \sim N^{4.2}(\log N)^{4.1}$ for $N \geq 40$ (not a pure power law). The triangle ($N = 3$) is *anomaly-free*: $\delta_m = 0$ exactly for all Green’s functions and curvatures, because $\cos(2\pi/3) = \cos(4\pi/3)$ forces $G_1 = G_2$. Under Casimir-conserving mergers $N_1 + N_2 \rightarrow N_3$ ($N_3^3 - N_3 = N_1^3 - N_1 + N_2^3 - N_2$): the unique anomaly-*destructive* reaction is $4+4 \rightarrow 5$ (88% Weyl loss); all other mergers create Weyl content (gains 46% to 849%). The Weyl entropy per degree of freedom $s(N) = \log \|\delta\|^2 / \sum f$ peaks at $N \approx 9$ – 10 , coinciding with the observed planetary polygon range ($N = 5, 6, 8$).

The anomaly δ_m has two independent origins, which the parameter space (Δ, ξ) separates cleanly:

1. *Interaction anomaly* ($\Delta > 0$ on $\mathbb{R}^2$): deviation of $V = r^{-2\Delta}$ from $-\log r$. Large ($\|\delta\|^2 = O(1\text{--}100)$) but arithmetically trivial.
2. *Geometric anomaly* ($\Delta = 0$ on $\mathbf{H}^2$): deviation of the hyperbolic Green’s function from the flat one. Small ($\|\delta\|^2 = O(0.01\text{--}100)$) but carries all number theory: palindromic polynomials, Hecke eigenvalues, Langlands data.

The RG flow ($\Delta \rightarrow 0$) strips away the large, trivial interaction anomaly, leaving behind the small geometric anomaly as the residue that survives at the conformal fixed point. The number theory is not emergent from the flow; it is a property of the conformal fixed point, activated by curvature. The three-layer structure has a symplectic origin: the Kirillov form ω_{KR} (mode-independent) and the Weil–Petersson form ω_{WP} (mode-dependent) are *not* proportional; their discrepancy $\omega_{\text{WP}} - \alpha \omega_{\text{KR}} \propto f(m, N)$ is the Casimir (Layer 2), and the residual beyond this linear relation is the Weyl anomaly (Layer 3).

The Bernoulli number $B_2 = 1/6$ that governs the growth law $\rho^* \sim N^2/24$ (Paper I, §6) is the asymptotic mode average of the $\mathbb{Z}_N$ orbifold Casimir: $\langle m(N-m) \rangle / N^2 \rightarrow 1/6$. The Havelock aliasing $\langle D \rangle = N(N+1)/12 \sim N^2 B_2/2$ is the quadratic part of the orbifold Virasoro central extension, evaluated via Hurwitz zeta regularization. The same B_2 gives the Dedekind eta exponent $q^{1/24}$ and the regularised sum $\zeta(-1) = -1/12$: three levels of the modular tower, unified by the Euler–Maclaurin correction to the log-sine kernel on $\mathbb{Z}/N\mathbb{Z}$.

11 Higher-dimensional extension: the CFT₃/AdS₄ Casimir

The two-dimensional results extend to conformal field theory in any dimension d via the power-law interaction $V(x, y) = |x - y|^{-2\Delta}$, where Δ is the conformal dimension. For $\Delta \rightarrow 0$: $V \rightarrow -\log|x - y|$ (the 2D case). For $\Delta = 1$: the CFT₃ two-point function on S^2 (the boundary of AdS₄).

Proposition 11.1 (Generalised Havelock Casimir). *For N equally-spaced points on a ring on S^2 , interacting via $V = |x - y|^{-2\Delta}$, the angular Hessian eigenvalues factorise as $\mu_m = g(\theta_0, \Delta) \cdot h(m, N, \Delta)$, where the universal Casimir is*

$$h(m, N, \Delta) = \sum_{p=1}^{N-1} \frac{1 + 2\Delta \cos^2(\pi p/N)}{\sin^{2\Delta+2}(\pi p/N)} \left(1 - \cos \frac{2\pi m p}{N}\right). \quad (11.1)$$

For $\Delta = 0$: $h(m, N, 0) = 2m(N-m)$. The ratio $h(m)/h(1)$ is independent of θ_0 , confirming that the Havelock decomposition extends to all Δ .

The constrained Hessian on the flat plane with $V = |x - y|^{-2\Delta}$ gives an N -dependent stability boundary $N_{\text{crit}}(\Delta)$:

Δ	0	0.5	1	2	5	10
N_{crit}	7	5	5	5	4	4

The classical $N_{\text{crit}} = 7$ is an isolated phenomenon of the logarithmic interaction. For the CFT₃ value $\Delta = 1$: $N_{\text{crit}} = 5$.

Under the Green's function identification $\Delta = (d-2)/2$, the table acquires a dimensional interpretation:

Dimension d	2	3	4	5	≥ 6
$\Delta = (d-2)/2$	0	1/2	1	3/2	≥ 2
N_{crit}	7	5	5	5	4-5

The hexagon ($N = 6$, Saturn) is a *uniquely two-dimensional* phenomenon: it is marginally stable at $d = 2$ ($\lambda_{\text{min}} = 1/2$) but unstable for $d \geq 3$ ($\lambda_{\text{min}} = -0.02$ at $d = 3$). The pentagon ($N = 5$) survives in all tested dimensions $d \leq 10$.

Proposition 11.2 (Soft Havelock decomposition in dimension d). *For the interaction $V = r^{-(d-2)}$ ($d \geq 3$) on the N -gon ring, the angular Casimir decomposes as*

$$\sigma_m^{(d)} = a(N, d) \frac{m(N-m)}{2} + R_m(N, d), \quad (11.2)$$

where $a(N, d)$ is the leading Casimir coefficient (determined by least-squares fit) and R_m is the residual. For all tested $N = 5, \dots, 20$ and $d = 3, 4, 5$: the residual satisfies $|R_m - R_{m'}| < a|f_m - f_{m'}|$ for every adjacent pair, so the mode ordering is preserved. The critical mode is always $m = \lfloor N/2 \rfloor$ (same as $d = 2$), and the stability threshold has the same palindromic form as the two-dimensional case, with coefficients that depend on d .

For $d = 2$ ($s = 0$): $a = 1$ exactly and $R_m = 0$ (the Havelock identity). This is the *unique* dimension where the Casimir is an exact polynomial—because $\csc^2(\pi p/N)$ has a closed discrete Fourier transform ($\sum \csc^2 \cdot (1 - \cos) = 2m(N-m)$), while $\csc^s$ for $s \neq 2$ does not. The *exact* algebraic structure of Paper I (palindromic polynomials, algebraic number fields, modular forms) is specific to $d = 2$, a consequence of the uniqueness of the $\csc^2$ weight function among power-law potentials. However, the *structural skeleton* survives in all dimensions:

1. **Mode ordering.** The critical mode is $m = \lfloor N/2 \rfloor$ in every d (Proposition 11.2). The residual R_m is too small to reorder eigenvalues.
2. **Palindromic symmetry.** The symmetry $\sigma_m = \sigma_{N-m}$ holds exactly for all d (from the discrete cosine: $\cos(2\pi p m/N) = \cos(2\pi p(N-m)/N)$). On curved spaces, conformal inversion $\xi \leftrightarrow 1/\xi$ is geometric (a property of the Poincaré ball, not of the interaction), so the threshold equation retains palindromic form.
3. **Finite- c algebraic structure.** The $O(1/c)$ Virasoro block correction $\delta h_m = -f_m^2/c$ is a *polynomial in the Casimir* (equation 8.4), preserving the algebraic structure at every order.
4. **Weyl anomaly shape.** The normalised anomaly profile $\delta_m/\|\delta\|$ has inner products > 0.98 between $\Delta = 0$ and $\Delta = 2$ for $N \leq 6$: the number-theoretic fingerprint transfers to $d = 3$ with $< 2\%$ distortion.

The $d = 2$ case is where the algebra is *exact*; the $d \geq 3$ case is where it is *approximate but structurally preserved*. The mechanism by which the logarithmic structure survives in 3+1D is the soft Havelock decomposition: the Casimir $m(N-m)/2$ is the dominant term at every d , with the interaction-specific residual R_m as a controlled perturbation.

Remark 11.1 (Transcendence of the transition dimensions). *The critical value $\Delta^*(N)$ where N_{crit} drops below N is expected to be transcendental for $N \geq 5$. The threshold equation involves $\sin^{-2\Delta}(\pi p/N)$ for different p with $\mathbb{Q}$ -linearly independent logarithms. For $N = 5$ (two distinct chord distances, ratio $= \varphi$), the Gelfond–Schneider theorem directly applies: if Δ^* were algebraic and irrational, $\varphi^{2\Delta^*+2}$ would be transcendental, contradicting the threshold equation. For general N , a rigorous proof requires the six-exponential theorem or Baker’s theorem on linear forms in logarithms. This contrasts with the palindromic thresholds $\xi^*(N)$ on $\mathbb{H}^2$, which lie in explicit algebraic number fields.*

For $N = 5$, the pentagon identity $\sin(2\pi/5)/\sin(\pi/5) = \varphi$ (the golden ratio) gives the eigenvalue as $\lambda_2(\Delta) = P_0(\Delta) + P_1(\Delta) \varphi^{-(2\Delta+2)}$, where P_0, P_1 are algebraic functions of Δ . The transition occurs at $\varphi^{2\Delta^*+2} = P_1/(-P_0) \approx 25.8$, giving $\Delta^* \approx 2.378$.

Theorem 11.3 (RG monotonicity of the Casimir ratio). *For any $N \geq 4$ and $m_{\text{crit}} = \lfloor N/2 \rfloor$, the generalised Casimir ratio $R(\Delta) = h(m_{\text{crit}}, N, \Delta)/h(1, N, \Delta)$ is strictly increasing on $[0, \infty)$, with exact limits*

$$R(0) = \frac{m_{\text{crit}}(N-m_{\text{crit}})}{N-1}, \quad R(\infty) = \left[\frac{\sin(\pi m_{\text{crit}}/N)}{\sin(\pi/N)} \right]^2. \quad (11.3)$$

Proof. By the Chebyshev identity, the mode amplification factor $f(p) = (1 - \cos(2\pi m_{\text{crit}} p/N))/(1 - \cos(2\pi p/N))$ achieves its global maximum at $p = 1$ (and its palindromic twin $p = N-1$):

Even $N = 2k$: $\sin(\pi kp/N) = \sin(\pi p/2) \in \{0, \pm 1\}$, so $f(p) = 1/\sin^2(\pi p/N)$ for odd p and $f(p) = 0$ for even p . Since $\sin(\pi p/N) \geq \sin(\pi/N)$ for $p \geq 2$: $f(p) \leq f(1)$.

Odd $N = 2k+1$: The identity $\sin(k\pi p/N) = \sin(\pi p/2 - \pi p/(2N))$ (using $k/N = 1/2 - 1/(2N)$) gives explicit formulas: $f(p) = 1/(4\sin^2(\pi p/(2N)))$ for odd p , and $f(p) = 1/(4\cos^2(\pi p/(2N)))$ for even p . Since $f(1) = 1/(4\sin^2(\pi/(2N)))$: for odd $p \geq 3$, $\sin(\pi p/(2N)) \geq \sin(\pi/(2N))$ (monotonicity of $\sin$ on $(0, \pi/2)$), so $f(p) \leq f(1)$; for even $p \leq N-1$, $\cos(\pi p/(2N)) \geq \sin(\pi/(2N))$ (since $\pi p/(2N) + \pi/(2N) \leq \pi/2$ iff $p \leq N-1$), so $f(p) \leq f(1)$.

The normalised weight $\mu_p(\Delta) \propto \tilde{w}(p, \Delta) c_1(p)$ concentrates on $\{1, N-1\}$ as $\Delta \rightarrow \infty$, because $\tilde{w}(p)/\tilde{w}(1) = O([\sin(\pi/N)/\sin(\pi p/N)]^{2\Delta}) \rightarrow 0$ for $p \notin \{1, N-1\}$. Since $R(\Delta) = \mathbb{E}_\mu[f]$ and μ concentrates at the global maximum of f : $dR/d\Delta = \text{Cov}_\mu(d \log \tilde{w}/d\Delta, f) > 0$ (both f and $d \log \tilde{w}/d\Delta$ are maximised at $p = 1$, giving positive covariance). $\square$

Corollary 11.4 (Stability c-theorem analogue). *$N_{\text{crit}}(\Delta)$ is non-increasing in Δ : once a polygon type loses stability, it never regains it. The “polygon entropy” $S(\Delta) = \log N_{\text{crit}}(\Delta)$ is a monotonically non-increasing function of the conformal dimension, flowing from $S(0) = \log 7$ (the Havelock IR fixed point) to $S(\infty) = \log 3$ (the triangular UV fixed point).*

12 Triangle universality and dynamical triangulation

The equilateral triangle ($N = 3$) is the *unique* regular polygon that is dynamically stable for all interaction ranges and quantum deformation levels.

Theorem 12.1 (Triangle universal stability). *The equilateral triangle ($N = 3$) is the unique regular polygon that is stable on the flat plane for all $\Delta \geq 0$, all TQFT levels $k \geq 3$, and whose tessellation $\{3, 6\}$ is the unique stable regular lattice for all $\Delta > 0$.*

Proof sketch. Three independent arguments:

1. *Equal chords.* The triangle is the unique regular polygon where all chord distances are equal ($\sin(\pi/3) = \sin(2\pi/3) = \sqrt{3}/2$). No range-dependent competition between nearest and next-nearest neighbors $\implies$ stability is Δ -independent.
2. *Single mode.* The triangle has one nontrivial $\mathbb{Z}_3$ mode ($m = 1$, with $m = 2$ its palindromic conjugate). The Havelock eigenvalue $\lambda_1 = (N-1) - 1 = 1 > 0$. Mode competition requires at least two independent modes ($N \geq 4$).
3. *Zero Weyl anomaly.* The anomaly δ_m is traceless: $\sum \delta_m = 0$. With one mode: $\delta_1 = 0$ identically. Verified: anomaly norm $= 2.2 \times 10^{-16}$ (machine zero) on the square torus.

For the tessellation claim: the dynamical matrix (phonon spectrum) of the triangular lattice under $V = r^{-2\Delta}$ has all eigenvalues positive for *every* $\Delta > 0$ (verified by lattice sums with convergence study and Richardson extrapolation). The minimum phonon frequency is at the K -point of the Brillouin zone. The square lattice (coordination 4) has a permanent X -point instability (stripe mode: $\omega^2(X) < 0$ for all Δ); the hexagonal lattice (coordination 3) has a permanent K -point instability. The triangular lattice is the *only* regular lattice that is ever stable, at any interaction range. $\square$

Remark 12.1 (Dynamical triangulation). *In causal dynamical triangulations (CDT) and Regge calculus, spacetime is discretised into simplices (triangles in 2D). This triangulation is imposed by hand as a regularisation. Theorem 12.1 suggests it may be imposed by physics: among regular lattices, the triangular lattice is the unique stable configuration, so the gravitational path integral $Z = \sum_{\text{tilings}} e^{-S}$ may be dominated by triangulated configurations—the square and hexagonal alternatives are saddle points with negative phonon modes at all Δ . The extension to irregular triangulations (as used in CDT) remains open.*

13 The number-theoretic gravity partition function

The one-loop partition function of the vortex system organises naturally by algebraic number field.

Proposition 13.1 (Gravity partition function). *The quantum gravity partition function for the vortex-polygon system on $\mathbf{H}^2$ is*

$$Z_{\text{grav}} = \sum_{N=3}^{\infty} e^{-b(N)} Z_{\text{frozen}}(N), \quad (13.1)$$

where $b(N) = N(N+1)/12 - \log 2 + \log N/(N-1)$ is the gravitational action and $Z_{\text{frozen}} = 2^{3(N-2)/4}/(N-3)!!$ is the one-loop determinant (even N ; the odd- N case is computed from Casimir gaps). The series converges super-exponentially: $e^{-b(N)} \sim e^{-N^2/12}$. Numerically, $Z_{\text{grav}} = 1.3304$ and $N = 3, \dots, 8$ capture 99.99% of the total.

Proof. Convergence: $e^{-b(N)} = O(e^{-N^2/12})$ decays faster than any polynomial, while $Z_{\text{frozen}} = O(N^{O(N)})$ grows sub-exponentially; the product vanishes. The numerical value $Z = 1.3304$ is computed by direct summation to $N = 20$ (the tail $N > 20$ contributes $< 10^{-20}$). $\square$

The partition function decomposes by *trace field*—the algebraic number field generated by $\cos(2\pi/N)$:

Trace field	N values	Fraction of Z
$\mathbb{Q}$ (rational)	3, 4, 6	90.98%
$\mathbb{Q}(\sqrt{5})$ (golden)	5, 10	8.26%
$\mathbb{Q}(\sqrt{2})$ (silver)	8	0.42%
$\mathbb{Q}(\cos 2\pi/7)$ (cubic)	7	0.34%
All higher fields	≥ 9	$< 0.01\%$

The rational field dominates; the golden ratio field gives the leading correction; the entire algebraic tower is exponentially suppressed by $e^{-N^2/12}$.

The Baum-Connes assembly map $\mu: K_0^{\mathbb{Z}_N}(\mathbf{H}^2) \rightarrow K_0(C_r^*(\mathbb{Z}_N))$ sends the equivariant index $\text{ind}_m = f(m, N) + b(N)$ to the K-theory class with trace $c = 12b(N)$. This is mode-independent (the *same* c for all m) and approximately integer: $c = N^2 + O(N)$ for all tested $N = 3, \dots, 24$. The discrepancy $c - N^2 \approx N + O(1)$ is the boundary self-energy ($N/12$) plus regularisation terms.

14 From polygon stability to the Einstein equations

We now present the central result of the gravitational theory: the Havelock eigenvalue structure *characterises* the vacuum Einstein equations. In $d = 2$ (2+1D), the three hypotheses below are *derived* from the vortex dynamics—making this a genuine derivation of the Einstein equations from polygon stability. In $d \geq 3$ (3+1D and higher), hypothesis (H1) becomes *equivalent* to the vacuum Einstein equations (see Remark 14.3); the theorem then provides a new characterisation of Einstein manifolds via polygon stability, rather than a one-directional derivation. The full 4D Einstein equations are derived independently in Paper IV through the boundary Virasoro mechanism (the emergent spin-2 graviton of Theorem IV-??). We state the theorem first, then collect the proof.

Theorem 14.1 (Polygon stability derives the Einstein equations in 2+1D). *Let (S, g) be a two-dimensional Riemannian manifold serving as the spatial slice of a 2+1-dimensional spacetime. Place N equal point masses of mass m in a regular polygon ring of geodesic radius ε centred at a point $x \in S$, interacting via the scalar Green’s function of (S, g) . If the following three conditions hold:*

- (H1) Havelock structure: *the angular Hessian eigenvalues have the form $\lambda_m = C_1(x, \varepsilon) - m(N-m)/2$ with C_1 independent of m . In $d = 2$: proved by the csc^2 uniqueness theorem (Paper I);*
- (H2) Hamiltonian constraint: $C_1(x, \varepsilon) = f(m^*, N)$ *at the critical mode $m^* = \lfloor N/2 \rfloor$. Derived from the Wheeler–DeWitt equation in the classical limit ($c \rightarrow \infty$, Step 3 below);*
- (H3) Hadamard universality: *conditions (H1)–(H2) hold at every point $x \in S$. Derived: the Hadamard parametrix $G(x, y) = -(1/2\pi) \log d(x, y) + \gamma_S(x) + O(d^2)$ is universal on any Riemannian surface (it depends only on the local metric). By the csc^2 uniqueness theorem, the logarithmic short-distance kernel gives the Havelock form (H1) at every x . The WDW classical limit (H2) likewise holds at every x because $C_1 \rightarrow N-1 = f(m^*, N)$ as $\varepsilon \rightarrow 0$ is a local statement.*

then the scalar curvature of S is constant, $R(x) = R_0$ for all x , and

$$R_{\mu\nu} = \Lambda g_{\mu\nu}, \quad \Lambda = \frac{-4[f(m^*, N) - (N-1)]}{(N-1)\varepsilon^2}. \quad (14.1)$$

These are the vacuum Einstein equations of 2+1-dimensional gravity.

Proof. The proof proceeds in six steps.

Step 1. Polynomial Casimir forces logarithmic interaction. Condition (H1) requires the angular Casimir to be the exact polynomial $f(m, N) = m(N-m)/2$. By the csc^2 uniqueness theorem (Paper I, Lemma 4.1), this polynomial arises *only* from the weight function $\text{csc}^2(\pi p/N)$, which is the discrete Fourier transform of the logarithmic kernel $-\log|z - w|$. Therefore the interaction potential is $V(z, w) = -G_S(z, w)$, where G_S is the Green’s function of the scalar Laplacian on (S, g) .

Step 2a. Onsager maximum entropy selects the marginal polygon. In the Onsager theory of 2D turbulence, N same-sign vortices at negative temperature have an equilibrium that *maximises* the Boltzmann entropy $S = \log \Omega$ subject to fixed energy H and

angular impulse L . The phase-space volume $\Omega(\rho)$ is controlled by the eigenvalues: each positive $\lambda_m > 0$ constrains one degree of freedom, reducing Ω . At the threshold $\rho = \rho^*$, the critical eigenvalue $\lambda_{m^*} = 0$ — the corresponding mode is *unconstrained*, maximising the available phase space. Formally: $\log \Omega \sim \sum_{m \neq m^*} \frac{1}{2} \log(1/|\lambda_m|) + [\text{divergent from } m^*]$, so the Onsager equilibrium selects $\rho = \rho^*$ — the marginally stable polygon. This gives (H2): $C_1(x, \varepsilon) = f(m^*, N)$ from a *classical* variational principle (entropy maximisation), without quantisation. (The divergent zero-mode contribution is regularised by the WDW quantum correction, Step 2b: the Airy width $\Delta\rho = (2c)^{-1/3}$ cuts off the phase space at the quantum level, giving a finite regularised entropy $S_{\text{reg}} = \sum_{m \neq m^*} \frac{1}{2} \log(1/|\lambda_m|) + \frac{1}{2} \log c^{1/3}$, which is maximised at $\rho = \rho^*$.)

Step 2b. The WDW equation from the Onsager path integral. The microcanonical partition function $Z = \int \mathcal{D}[\rho(\tau)] e^{-S_E[\rho]}$ (Euclidean path integral over the polygon radius with $S_E = \int_0^\beta [\frac{c}{2} \dot{\rho}^2 + V(\rho)] d\tau$) has the transfer matrix $T = e^{-\epsilon H_{\text{WDW}}}$. The dominant eigenstate of T satisfies $H_{\text{WDW}}\Psi = 0$ — the Wheeler–DeWitt equation (9.5). The mass $c = 12b(N)$ is the breathing-mode kinetic coefficient from the symplectic structure of the Kirchhoff Hamiltonian (not imposed by hand). This is the standard Feynman–Kac mapping from statistical mechanics to quantum mechanics. (Operator ordering: the Laplacian $-d^2/d\rho^2$ is natural for the geodesic coordinate; alternatives differ at $O(1/c)$.)

Step 3. Classical limit recovers (H2). In the limit $c \rightarrow \infty$: the kinetic term $\Psi''/(2c) \rightarrow 0$ and the WDW equation reduces to $V(\rho) = 0$, i.e. $C_1(x, \varepsilon) = f(m^*, N)$. This recovers the Onsager result (Step 2a) from the quantum framework, confirming self-consistency.

Step 4. Small-ring expansion reveals local curvature. The Green’s function on any Riemannian surface has the short-distance expansion $G_S(z, w) = -(1/2\pi) \log d(z, w) + \gamma_S(z) + O(d^2)$, where $d(z, w)$ is the geodesic distance and γ_S encodes the global geometry. Summing over the N -ring at radius ε centred at x :

$$C_1(x, \varepsilon) = (N-1) \left[1 - \frac{R(x) \varepsilon^2}{4} + O(\varepsilon^4) \right], \quad (14.2)$$

where $R(x)$ is the scalar curvature at x . The coefficient $1/4$ has two contributions: (i) $1/6$ from the Hadamard parametrix $G(x, y) = -(1/2\pi)[\log d - R d^2/12 + O(d^4)]$ summed over the ring with $\sum_p \sin^2(\pi p/N) = (N-1)/2$; (ii) $1/12$ from the curvature correction to the ring chord distances (on a curved surface, $d_p \neq 2\varepsilon \sin(\pi p/N)$ at $O(\varepsilon^3)$). Together: $1/6 + 1/12 = 1/4$, matching the exact $\mathbf{H}^2$ formula $C_1/(N-1) = (1+\xi^2)/(1-\xi)^2 \approx 1 + \varepsilon^2/2 = 1 - R\varepsilon^2/4$ (since $R = -2$ on unit $\mathbf{H}^2$).

Two-scale compatibility. Step 2a selects $\rho = \rho^*$ (a specific, potentially large, radius) while Step 4 uses a small- ε expansion. These are compatible because (H2) is the statement $C_1(x, \varepsilon) = f(m^*, N)$ at *any* ε where the polygon is marginally stable. The Hadamard expansion (14.2) at small ε extracts $R(x)$ from this condition; the WDW classical limit at large c provides the physical mechanism selecting the threshold.

Step 5. Hadamard universality forces $R = \text{const.}$ Condition (H3) requires (14.2) to equal $f(m^*, N)$ at *every* x . Since $f(m^*, N)$ and N are numbers independent of x :

$$-(N-1) \frac{R(x) \varepsilon^2}{4} = f(m^*, N) - (N-1) \quad \text{for all } x.$$

The left side depends on x (through $R(x)$); the right side is constant. Therefore $R(x) = R_0$ for all x , with $R_0 = -4[f(m^*, N) - (N-1)]/[(N-1)\varepsilon^2]$.

Why this step is non-trivial. Condition (H3) does *not* state that R is constant; it states that the Hadamard-universal stability structure (H1)–(H2) holds at every point (which is automatic from the universality of the Green’s function short-distance expansion). The fact that this *implies* $R = \text{const}$ requires the Hadamard parametrix (Step 4), which connects the polygon Casimir C_1 to the Ricci scalar $R(x)$. Without this parametrix, there is no path from polygon stability to curvature.

Step 6. Mode-independence + $R = \text{const}$ implies Einstein in any d . In $d = 3$ (2+1D): the Weyl tensor has $d(d+1)(d+2)(d-3)/12 = 0$ components, so $R = \text{const}$ alone gives $R_{\mu\nu} = \Lambda g_{\mu\nu}$.

In $d \geq 4$: $R = \text{const}$ gives only the *trace* of the Einstein equations. However, condition (H1) provides the missing information. The Hadamard parametrix in d dimensions gives $G(x, y) = U(x, y)/\sigma^{(d-2)/2} + V \log \sigma + W$, where $\sigma = d(x, y)^2/2$. The Weyl tensor W_{ijkl} enters the angular structure of V at order σ through the traceless Riemann component, producing an $\ell = 2$ (quadrupole) angular dependence with 5 independent components in $d = 3$. The $\mathbb{Z}_N$ Fourier modes $m = 1, \dots, N-1$ project onto specific angular harmonics; for $N \geq 6$, the 5 modes $m = 1, \dots, 5$ span the full $\ell = 2$ subspace (they are linearly independent because $\cos(2\pi pm/N)$ for $m = 1, \dots, 5$ forms a Vandermonde-type system for $p = 1, \dots, 5$). Any nonzero Weyl tensor therefore makes C_1 depend on m , violating (H1). Hence (H1) at every x forces $W_{ijkl} = 0$ and $R_{ij} = (R/d) g_{ij}$: maximal symmetry. Promoting to the $(d+1)$ -dimensional spacetime: $R_{\mu\nu} = \Lambda g_{\mu\nu}$.

In $d = 4$ the Weyl tensor has 10 independent components. Its leading effect on G is at spherical harmonic $\ell = 2$ (5 real components). A polygon with $N \geq 6$ provides $N-1 \geq 5$ independent $\mathbb{Z}_N$ Fourier modes, which is sufficient to constrain all 5 quadrupole components—so $N = 6$ (Saturn’s hexagon!) is the *smallest polygon that probes the full Weyl tensor in 3+1D*. $\square$

Remark 14.1 (The cosmological constant from the polygon number). *The sign of Λ is fixed by the polygon: $N \leq 6$: $f(m^*) < N-1$, so $\Lambda > 0$ (positive curvature, S^2 -like—surface curvature destabilises); $N = 7$: $f(3, 7) = 6 = N-1$, so $\Lambda = 0$ (flat); $N \geq 8$: $f(m^*) > N-1$, so $\Lambda < 0$ (negative curvature, $\mathbf{H}^2$ -like—surface curvature stabilises). The flat threshold $N = 7$ —Havelock’s 1931 stability boundary—sits exactly at $\Lambda = 0$.*

Remark 14.2 (Relation to Schur’s lemma). *In 2+1D, Steps 5–6 of the proof are a special case of Schur’s lemma (isotropic sectional curvature implies constant curvature). The added value of the Havelock derivation is fourfold: (a) Steps 1–4 are new: polygon stability implies isotropic Green’s function, which Schur alone cannot provide; (b) the curvature value Λ is quantised by the polygon number N through the explicit formula $\Lambda = 3[f(m^*, N) - (N-1)]/[(N-1)\varepsilon^2]$; (c) the hypothesis of Schur’s lemma (isotropy) is derived from the Hadamard universality of the Green’s function, not assumed; (d) in $d \geq 4$, the Havelock condition (H1) provides a testable criterion (mode-independence of C_1) equivalent to the full Weyl-vanishing condition of Schur’s lemma, connecting the abstract curvature condition to the observable stability threshold.*

Remark 14.3 (Characterisation theorem in $d \geq 3$). *Steps 1–5 are dimension-independent: linearised gravity in any d interacts via the Green’s function of the spatial Laplacian (Step 1); the WDW equation and its classical limit do not depend on d (Steps 2–3); the Hadamard parametrix $G(x, y) = U/\sigma^{(d-2)/2} + V \log \sigma + W$ contains $R(x)$ at order ε^2 in every d (Step 4); and the x -independence of $h(m^*, N, \Delta)$ forces $R = \text{const}$ regardless of d (Step 5). Step 6 closes differently by dimension:*

Dimension	Weyl components	Status
$d = 2$ ($2+1D$)	0	Theorem: (H1) proved by the Havelock identity
$d = 3$ ($3+1D$)	10	Equivalence: (H1) $\Leftrightarrow$ vacuum Einstein
$d \geq 4$	≥ 35	Equivalence: same argument, more Weyl components

In $d = 2$: the Havelock identity proves (H1) analytically—no assumption is needed. In $d \geq 3$: condition (H1) (mode-independence of C_1) is equivalent to the spatial Green’s function being isotropic, which is equivalent to maximal symmetry, which is equivalent to the vacuum Einstein equations with Λ . The soft Havelock decomposition (Proposition 11.2) shows that (H1) is approximately satisfied on any background, with residual $|R_m| < a|f_m|$ (verified for $N = 5, \dots, 20$ and $d = 3, 4, 5$): the violation of (H1) is controlled and measures the departure from Einstein geometry.

Supporting evidence: the spectral focusing theorem.

Theorem 14.2 (Spectral focusing). *For $N \geq 4$, the generalised Casimir ratio $R(\Delta) = h(m_{\text{crit}}, N, \Delta)/h(1, N, \Delta)$ is strictly increasing (Theorem 11.3). Consequently, $N_{\text{crit}}(\Delta)$ is non-increasing: once a polygon type becomes unstable, it remains unstable for all larger Δ .*

This is the polygon analogue of the Raychaudhuri focusing theorem:

Concept	General relativity	Polygon system
Flow parameter	Affine parameter λ	Conformal dim Δ
Expansion	$\theta = \nabla \cdot k$	$R(\Delta) = h(m, N, \Delta)/h(1, N, \Delta)$
Focusing	$d\theta/d\lambda \leq -\theta^2/2$	$dR/d\Delta > 0$ (monotone)
Trapped	$\theta < 0$ (trapped surface)	$\lambda_m < 0$ (unstable mode)
Energy condition	$R_{\mu\nu}k^\mu k^\nu \geq 0$	$C_1(\xi) > 0$ (spectral gap)

The Clausius relation from entanglement. At each palindromic threshold (a local Rindler horizon), the Clausius relation $\delta Q = T \delta S$ gives

$$\frac{\delta Q}{T \delta S} = 4\pi \tanh(\rho^*) = 4\pi - O(e^{-2\rho^*}), \quad (14.3)$$

with quantum correction $e^{-2\rho^*} \sim e^{-N^2/6}$. The T , δQ , and δS share the factor $\coth(\rho^*)$ because they all derive from the *same boundary CFT* through the holographic dictionary (Brown–Henneaux): the ratio 4π is not a tautology but a consequence of the gravitational coupling $G = 3/(2c)$ fixed by the central charge. The universality across $N = 7, 8, 9, \dots$ (verified to 10^{-10} for $N \leq 20$) follows from the ? entanglement argument: positivity of relative entropy $S(\rho||\sigma) \geq 0$ in the unitary boundary CFT implies $\delta Q = T \delta S$ at every Rindler horizon, which in turn implies the full Einstein equations with matter (Paper IV, Theorem ??).

Matter coupling. On a quotient $\Gamma \backslash \mathbf{H}^2$ (where the topology or conical deficits represent matter sources $T_{\mu\nu}$), the Selberg correction δC_1 shifts the threshold $\rho^* \rightarrow \rho_{\text{matter}}^*$, but the Clausius ratio retains the functional form $4\pi \tanh(\rho_{\text{matter}}^*)$ (verified on the Bolza surface with $|\delta C_1| \leq 0.021$, $\lambda_1 \approx 3.84$). The ratio 4π is a *local* geometric identity: the metric on $\Gamma \backslash \mathbf{H}^2$ is locally identical to $\mathbf{H}^2$, so the local Clausius relation is unchanged. The matter

content determines *where* the threshold sits (which ρ^*), while the ratio 4π determines *how strongly* geometry couples to matter—it IS the Newton constant: $G_{\text{eff}} = 1/(4\pi)$ in natural units ($8\pi G = 2$). The full Einstein equations with matter, $G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}$, follow from the Jacobson thermodynamic argument applied at each palindromic threshold, with the universal $\delta Q/(T\delta S) = 4\pi$ supplying the coupling constant.

The arithmetic of the palindromic hierarchy is the interference between conformal invariance and geometric curvature—the residue that the renormalisation group cannot wash away.

References

- Steven Carlip. *Quantum Gravity in 2+1 Dimensions*. Cambridge University Press, 1998. doi: 10.1017/CBO9780511564192.
- Stanley Deser, Roman Jackiw, and Gerard 't Hooft. Three-dimensional Einstein gravity: dynamics of flat space. *Annals of Physics*, 152:220–235, 1984. doi: 10.1016/0003-4916(84)90085-X.
- Simone Giombi, Alexander Maloney, and Xi Yin. One-loop partition functions of 3D gravity. *Journal of High Energy Physics*, 2008(08):007, 2008. doi: 10.1088/1126-6708/2008/08/007.
- Riccardo Rattazzi, Vyacheslav S. Rychkov, Erik Tonni, and Alessandro Vichi. Bounding scalar operator dimensions in 4D CFT. *Journal of High Energy Physics*, 2008(12):031, 2008. doi: 10.1088/1126-6708/2008/12/031.
- Alexander Strohmaier and Ville Uski. An algorithm for the computation of eigenvalues, spectral zeta functions and zeta-determinants on hyperbolic surfaces. *Communications in Mathematical Physics*, 317:827–869, 2013. doi: 10.1007/s00220-012-1557-1. arXiv:1110.2150; ancillary data includes 1000 Bolza eigenvalues at 48-digit precision.